# Harshith Reddy Nalla

Harshith.com — linkedin:// Harshith/ — harshithreddy0117@gmail.com —South Dakota, USA


## PROFESSIONAL SUMMARY


AI Engineer specializing in **deep learning** for **medical imaging** and computer vision, with proven experience developing production-ready models using **PyTorch** and **TensorFlow**, published research at **AAAI 2026**, and delivering scalable AI systems that achieve **92%+ accuracy** in clinical diagnostics.


## EXPERIENCE

**Research Assistant** 06/2025 – Present

*Computer Science Department, University of South Dakota*

- Developed deep-learning models (**CNNs, Transformers, Mamba** architectures) for medical diagnostics in pancreatic cancer and dental imaging, achieving **92% accuracy** in early detection and improving diagnostic precision by **25%** compared to baseline models through advanced data preprocessing and training pipeline optimization.
- Built and deployed full-stack medical AI systems using React, Spring Boot, **FastAPI**, **Docker**, and **Hugging Face**, enabling **real-time inference** on 500+ medical scans with sub-2-second response times, 3D visualization capabilities, and automated report generation that reduced radiologist review time by **40%**.
- Improved model accuracy and reliability by implementing enhanced data preprocessing pipelines, custom training strategies, and comprehensive evaluation metrics (**Dice score, IoU**, sensitivity/specificity), making research results publication-ready and contributing to accepted paper at **AAAI 2026 Workshop**.
- Conducted extensive benchmarking experiments comparing CNN, Transformer, and Mamba architectures across **1,000+ medical imaging cases**, performed rigorous result analysis, and contributed to scientific writing for research papers demonstrating CNN superiority in dental caries segmentation tasks.
- Collaborated with medical researchers and clinicians to translate research requirements into technical specifications, iteratively refined models based on clinical feedback, and delivered AI tools that directly improved early cancer detection workflows in real clinical settings.

**Student Government Association** 08/2023 – 08/2024

*Secretary of Finance Department, University of South Dakota*

- Elected to SGA senate through campus-wide voting; managed allocation of $200,000+ annual student activity budget across 50+ campus organizations and events, improving fund distribution efficiency by 35% through systematic process improvements.
- Collaborated with cross-functional teams including Revenue Operations, Marketing, and executive leadership to address student concerns, improve campus life initiatives, and ensure transparent financial decision-making while maintaining strict confidentiality of sensitive budget information.
- Represented 8,000+ student interests on key campus issues, volunteered 100+ hours in office hours and senate meetings, and demonstrated leadership in policy decisions affecting student programming, facility improvements, and organizational funding priorities.

## PROJECTS

**Liver Profile AI** - Developed AI-based system using **PyTorch**, **3D CNNs**, and **Mamba** architectures for automated liver segmentation and anomaly detection from CT/MRI scans. Achieved **88% Dice score** on LiTS dataset, integrated model into FastAPI backend with real-time visualization, and deployed on **Hugging Face Spaces** with interactive overlays for clinical interpretation. Implemented efficient inference pipeline processing 200+ scans with **95% segmentation accuracy**.

**DentiMap** - Full-stack web application using **Python**, **PyTorch**, and **CNNs** to analyze dental X-rays for AI-driven diagnostics. Built with Java Spring Boot, FastAPI backend and responsive React frontend, achieved **87% accuracy** in caries detection, and delivered instant results through clean, user-friendly interface processing diagnostic cases.

## TECHNICAL SKILLS

---

- **Languages:** Python, Java, TypeScript, JavaScript, SQL
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV
- **Deep Learning:** CNNs, Transformers, Mamba Architectures, Computer Vision, Medical Image Analysis, Model Optimization
- **Backend Development:** FastAPI, Spring Boot, Spring Framework, RESTful APIs, Microservices
- **Frontend Development:** React, TypeScript, Tailwind CSS, Responsive Design
- **DevOps & Deployment:** Docker, Kubernetes, AWS (Pipeline, ECS, EKS, EC2, S3), Hugging Face Spaces, CI/CD
- **Testing & Tools:** Git & GitHub, Model Evaluation Metrics (Dice Score, IoU, F1)
- **Databases:** SQL Server, PostgreSQL, MySQL

## PUBLICATIONS

---

**When CNNs Outperform Transformers and Mambas: Revisiting Deep Architectures for Dental Caries Segmentation**
Aashish Ghimire, Jun Zeng, Roshan Paudel, Nikhil Kumar Tomar, Deepak Ranjan Nayak, **Harshith Reddy Nalla**, Vivek Jha, Glenda Reynolds, Debesh Jha
AAAI 2026 Workshop — arXiv: https://arxiv.org/pdf/2511.14860, Nov 2025

## EDUCATION

---

**University of South Dakota**
BS, Computer Science - AI — 01/2023 – 05/2027
GPA: 3.6 — Rising Senior

**DSA with Java**
Professional Certificate — 05/2025